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#DS_Practical_4
#Pythan Language
class Student:
        def __init__(self, roll, name, marks):
        self.roll = roll
        self.name = name
        self.marks = marks
        self.next = None



class StudentRecord:
        def __init__(self):
        self.head = None

        def add_student(self):
        roll = int(input("Enter Roll Number: "))
        name = input("Enter Name: ")
        marks = float(input("Enter Marks: "))

        new_node = Student(roll, name, marks)

        if self.head is None:
         self.head = new_node
        else:
        temp = self.head
        while temp.next is not None:
                temp = temp.next
        temp.next = new_node
```

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print("Student added successfully!")

def display(self):
if self.head is None:
print("No student records found.")
return

temp = self.head
print("\nRoll No.\tName\t\tMarks")

while temp is not None:
print(temp.roll, "\t\t", temp.name,
      "\t\t", temp.marks)
temp = temp.next

def search_student(self):
roll = int(input("Enter Roll Number to search: "))
temp = self.head

while temp is not None:
if temp.roll == roll:
        print("Student Found!")
        print("Roll Number:", temp.roll)
        print("Name:", temp.name)
        print("Marks:", temp.marks)
        return
temp = temp.next

print("Student not found.")
```

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def update_student(self):
roll = int(input("Enter Roll Number to update: "))
temp = self.head

while temp is not None:
if temp.roll == roll:
        temp.name = input("Enter New Name: ")
        temp.marks = float(input("Enter New Marks: "))
        print("Record updated successfully!")
        return
temp = temp.next

print("Student not found.")

def delete_student(self):
roll = int(input("Enter Roll Number to delete: "))
temp = self.head
prev = None

while temp is not None:
if temp.roll == roll:
        if prev is None:
        self.head = temp.next
        else:
        prev.next = temp.next

        print("Record deleted successfully!")
        return
```

```
prev = temp
temp = temp.next

print("Student not found.")

def sort_students(self):
if self.head is None or self.head.next is None:
print("Not enough records to sort.")
return

print("1. Sort by Roll Number")
print("2. Sort by Marks")
choice = int(input("Enter choice: "))

print("1. Ascending")
print("2. Descending")
order = int(input("Enter order: "))

if choice not in (1, 2) or order not in (1, 2):
print("Invalid choice.")
return

i = self.head
while i is not None:
j = i.next

while j is not None:
        if choice == 1:
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                a, b = i.roll, j.roll
                else:
                a, b = i.marks, j.marks

                if (order == 1 and a > b) or \
                (order == 2 and a < b):
                i.roll, j.roll = j.roll, i.roll
                i.name, j.name = j.name, i.name
                i.marks, j.marks = j.marks, i.marks

                j = j.next
        i = i.next

        print("Records sorted successfully!")
        self.display()


def main():
        records = StudentRecord()

        while True:
        print("\n--- Student Record Management ---")
        print("1. Add Student")
        print("2. Delete Student")
        print("3. Update Student")
        print("4. Search Student")
        print("5. Sort Records")
        print("6. Display Records")
        print("7. Exit")
```

```
        choice = input("Enter your choice: ")

        if choice == '1':
        records.add_student()
        elif choice == '2':
        records.delete_student()
        elif choice == '3':
        records.update_student()
        elif choice == '4':
        records.search_student()
        elif choice == '5':
        records.sort_students()
        elif choice == '6':
        records.display()
        elif choice == '7':
        print("Program terminated.")
        break
        else:
        print("Invalid choice. Try again.")


if __name__ == "__main__":
        main()
```

Sample Output:

```
--- Student Record Management ---
1. Add Student
2. Delete Student
```

3. Update Student

4. Search Student

5. Sort Records

6. Display Records

7. Exit

Enter your choice: 1

Enter Roll Number: 12

Enter Name: Vaishnavi

Enter Marks: 85

Student added successfully!

Enter your choice: 6

Roll No. Name Marks

12 Vaishnavi 85.0

Enter your choice: 7

Program terminated.